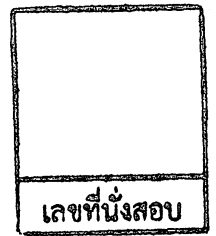


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2017

Subject: 341152 Electric Circuits II

Section: 5-6

Date: 9 March 2018

Time: 15.00-17.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. The examination has 13 pages (including this page), 3 parts (50 questions) and a total score of 70 points.
 2. Write all your solutions and answers on this examination sheet.
 3. This is a closed book examination.
 4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
 5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
 6. You are not allowed to use the restroom during the exam except in case of an emergency.
 7. No documents are allowed to be taken out of the examination room.
 8. Calculators are allowed in the examination.
 9. Electronic communication devices are NOT allowed in the examination room.
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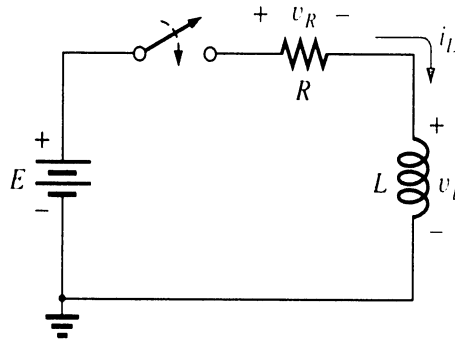
**Cheating in the exam is considered an extremely serious offence
which will result in expulsion from the University.**

Part 1 (20 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Capacitance is inversely proportional to the area of plates.
2. When charging capacitor, the current of capacitor decreases exponentially.
3. The transient state of the capacitor current can be found by substituting the short-circuit equivalent.
4. In RL network, the energy will be fully stored after 5τ .
5. In RC network, the steady state is occurred when the switch is turned on before τ .
6. When charging capacitor, the voltage of capacitor increases exponentially.
7. When closing switch of RC DC circuit, electric current can flow in right at the beginning.
8. The time constant for a capacitor circuit equals its capacitance multiplied by its total resistance.
9. An ideal inductor looks like a short circuit to dc current.
10. The current across a inductor leads the voltage through it by 90° .
11. The time constant for an inductor circuit equals its resistance divided by its inductance.
12. Power factor of resistive circuit is equal to zero.
13. Consider total power equation $P_T = EI \cos \theta_T$, the θ_T denotes the phase different between E and I .
14. The magnitude of polar form is in effective or RMS value.

15. Resistance value of resistor in ideal case depends on frequency.
16. The unit of average power is in Watt.
17. The unit of power factor $\cos(\theta)$ is in radius.
18. From circuit below, the voltage divider rule can be applied to calculate V_L .



19. The average value of the sinusoidal waveform is zero.
20. Given $Z_1 = 5\Omega$ and $Z_2 = j5\Omega$. If Z_1 and Z_2 is connected in the series configuration.
Then, $Z_T = 5 + j10\Omega$.

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Part 2 (25 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. What kind of capacitor is figure 1?

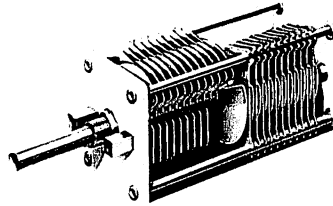


Figure 1

- (a) Ceramic trimmer capacitor (b) Air trimmer capacitor
(c) Mica capacitors (d) Polypropylene capacitors

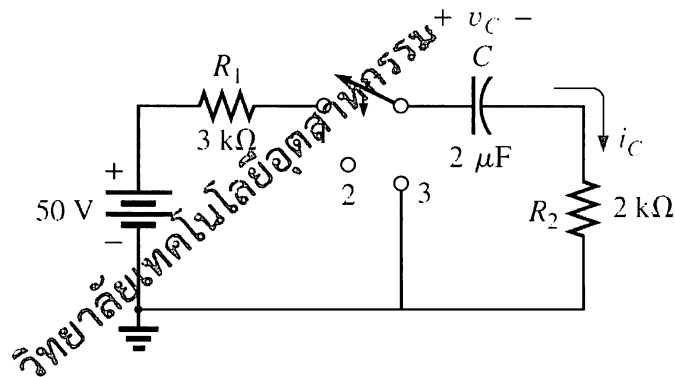


Figure 2

2. What is the time constant in the discharging state of the circuit from figure 2?

- (a) 10 ms (b) 4 ms (c) 10 s (d) 6 μs

3. What is the time constant in the charging state of the circuit from figure 2?

- (a) 10 ms (b) 4 ms (c) 10 s (d) 6 μs

4. What is the capacitor voltage at charging state in figure 2?

- (a) $50e^{t/\tau}$ [V] (b) $50e^{-t/\tau}$ [V] (c) $50(1 - e^{t/\tau})$ [V] (d) $50(1 - e^{-t/\tau})$ [V]

5. What is the capacitor current at charging state in figure 2?

- (a) $10e^{t/\tau}$ [mA] (b) $10e^{-t/\tau}$ [mA] (c) $5e^{t/\tau}$ [mA] (d) $5e^{-t/\tau}$ [mA]

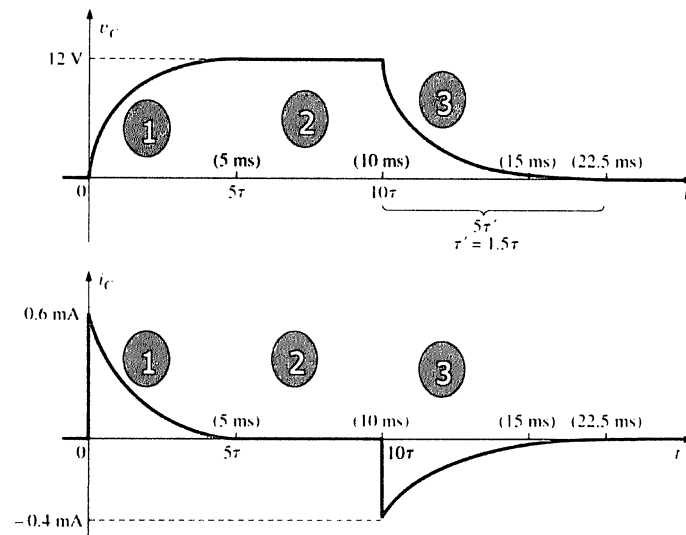


FIGURE 3

6. Find the time constant in region 1 of R-C circuit from figure 3.
 - (a) 0.1ms
 - (b) 0.5ms
 - (c) 1ms
 - (d) 1.5 ms
7. Find the voltage equation of capacitor in region 1 from figure 3.
 - (a) $12(1 - e^{-t/\tau})$
 - (b) $15(1 - e^{-t/\tau})$
 - (c) $12(1 + e^{-t/\tau})$
 - (d) $15(1 + e^{-t/\tau})$
8. Find the voltage equation of capacitor in region 2 from figure 3.
 - (a) 12 V
 - (b) 15 V
 - (c) $12(1 + e^{-t/\tau})$
 - (d) $15(1 + e^{-t/\tau})$
9. Find the capacitance of a parallel-plate capacitor if the area of each plate is 0.075 m^2 and the distance between the plates is 1.77 mm. The dielectric is paraffin paper ($\epsilon_r = 2.5$) where $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$.
 - (a) 0.522 pF
 - (b) 522.45 pF
 - (c) 0.937 pF
 - (d) 938.05 pF
10. What is the unit of inductance?
 - (a) Ohm
 - (b) Farad
 - (c) Tesla
 - (d) Henry

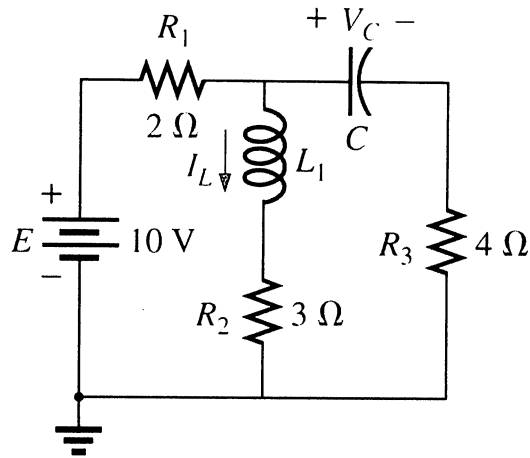


FIGURE 4

11. In steady state, what is the time constant of the circuit from figure 4?
 (a) 2.7 s (b) 0.0 s (c) 11 s (d) 11 ms
12. In steady state, what is the inductor current of the circuit from figure 4?
 (a) 1A (b) 2A (c) 4 A (d) 5A
13. In steady state, what is the inductor voltage of the circuit from figure 4?
 (a) 12 (b) 1.2V (c) 0V (d) 5V
14. In steady state, what is the capacitor voltage of the circuit from figure 4?
 (a) 12 (b) 6 (c) 0V (d) 5V
15. From table and circuit below, find the voltage across inductor at steady state ($t = 100\tau$). Given $E=20V$, $L=1mH$ and $R=100\Omega$.

TRANSIENT STATE		STEADY STATE
$t = 0$	$0 < t < 5\tau$	$t > 5\tau$

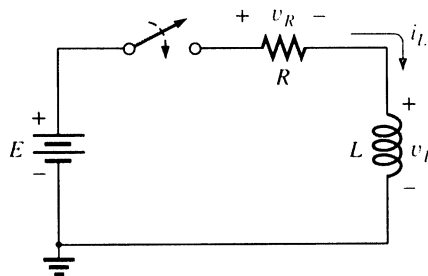


FIGURE 5

- (a) 0 V (b) 3 (c) 5 A (d) 7.5 A

16. From question 20, find the energy store in inductor.

- (a) 0 mJ (b) 0.005 mJ (c) 0.5 mJ (d) 5 mJ

17. What is the time domain equation of voltage (v) and current (i) from figure 6?

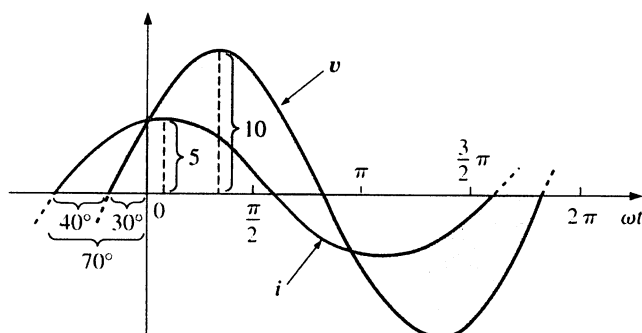


FIGURE 6

(a) $v = 10\sin(\omega t + 30^\circ)$, $i = 5\sin(\omega t + 70^\circ)$

(b) $v = 10\sin(\omega t + 30^\circ)$, $i = 5\sin(\omega t + 40^\circ)$

(c) $v = 10\sin(\omega t + 40^\circ)$, $i = 5\sin(\omega t + 70^\circ)$

(d) $v = 10\sin(\omega t + 0^\circ)$, $i = 5\sin(\omega t + 30^\circ)$

18. Find the capacitance of a parallel-plate air dielectric capacitor if the area of each of its plates is 0.1 m^2 and the distance between the plates is 2.0 mm.

- (a) 1.44 nF (b) 0.62 nF (c) 0.44 nF (d) 0.14 nF

19. What is the inductive reactance at 400 Hz of a 1 mH inductor?

- (a) 12Ω (b) 5.0Ω (c) 2.5Ω (d) 0.2Ω

20. What is the phasor form of voltage (v) in figure 7?

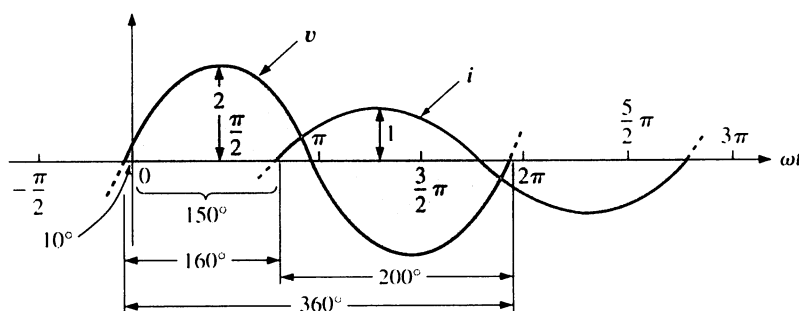


FIGURE 7

(a) $V = \frac{2}{\sqrt{2}} \angle 150^\circ$ (b) $V = \frac{2}{\sqrt{2}} \angle 10^\circ$ (c) $V = \frac{1}{\sqrt{2}} \angle 150^\circ$ (d) $V = \frac{1}{\sqrt{2}} \angle 10^\circ$

21. What is the phasor form of current (I) in figure 7?

- (a) $I = \frac{2}{\sqrt{2}} \angle 150^\circ$ (b) $I = \frac{2}{\sqrt{2}} \angle 10^\circ$ (c) $I = \frac{1}{\sqrt{2}} \angle -150^\circ$ (d) $I = \frac{1}{\sqrt{2}} \angle -10^\circ$

22. Find effective or rms value of signal in figure 8.

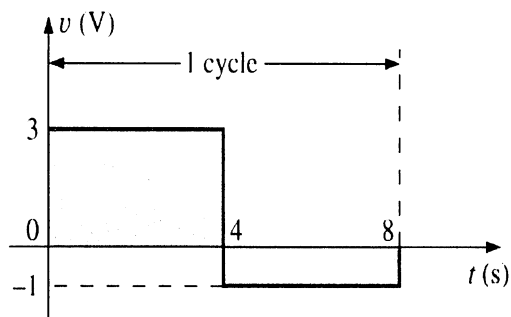


FIGURE 8

- (a) 2.0V (b) 2.24V (c) 2.54V (d) 3.2V

23. The voltage across a 100 mH coil is $v = 50 \sin 50t$. Which of these expressions describes the current?

- (a) $1000 \sin(50t - 90^\circ)$ (b) $10 \sin(50t + 90^\circ)$
(c) $10 \sin 50t$ (d) $10 \sin(50t - 90^\circ)$

24. Which one of the following values is equivalent to $1 \angle 180^\circ$?

- (a) $-1 \angle 0^\circ$ (b) $-1 \angle 90^\circ$ (c) $1 \angle 360^\circ$ (d) $1 \angle -90^\circ$

25. From circuit in Figure 9 find total current.

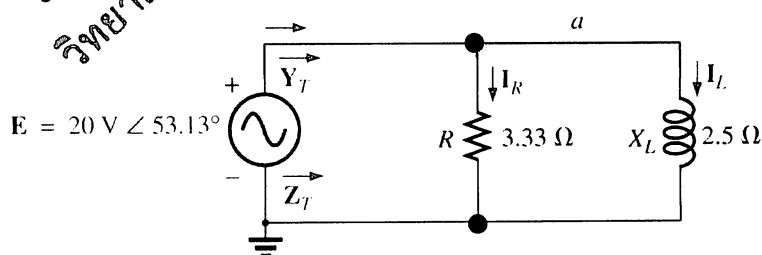


FIGURE 9

- (a) $10 \angle 0^\circ$ A (b) $10 \angle 15^\circ$ A (c) $50 \angle 180^\circ$ A (d) $10 \angle -15^\circ$ A

Part 3 (25 points) Calculation

Instruction: Show the mathematical expression and answer of following problems.

- Use figure 10 to find amplitude of capacitor voltage V_c and current I_c at 1τ to 6τ of the charging state and time constant. Given $E=20\text{ V}$, $R=10\text{ k}\Omega$ and $C=0.1\mu\text{F}$. Finally, draw the graph of V_c and current I_c time responses.

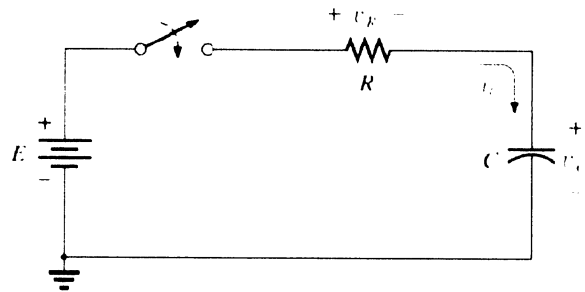


Figure 10

$\tau =$

V_c and I_c at $1\tau =$

V_c and I_c at $2\tau =$

V_c and I_c at $3\tau =$

V_c and I_c at $4\tau =$

V_c and I_c at $5\tau =$

V_c and I_c at $6\tau =$

2. Calculate the currents I_1 and I_2 of Figure 11 in phasor form using the current divider rule.

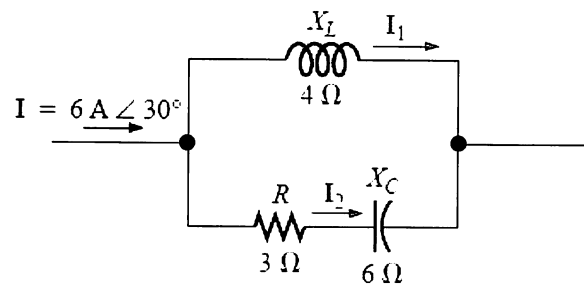


Figure 11

3. Find the total ADMITTANCE (Y_T) and IMPEDANCE (Z_T) of the circuit of Figure 12. And, find total current. Plot the Phasor diagram of all currents.

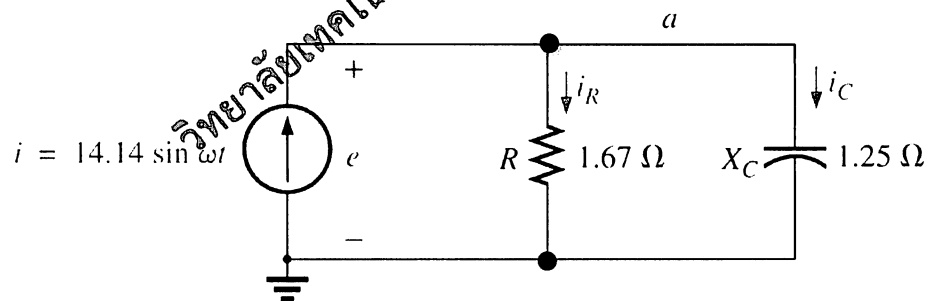


Figure 12

4. Use figure 13 to find total current I , I_R , I_L and I_C . Plot the Phasor diagram of all currents.

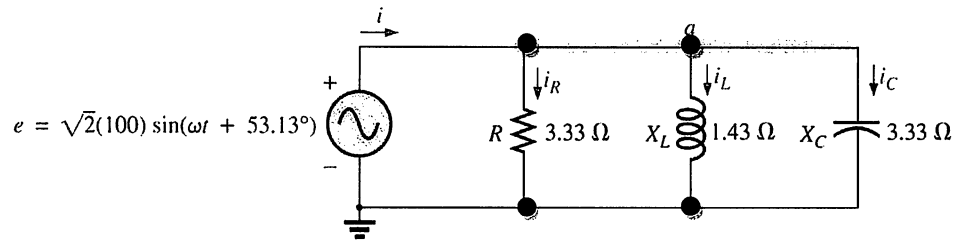


Figure 13

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5. In the circuit for Figure 14, find V_L and V_C . Given source $I = 2\sin(\omega t + 30^\circ)$ V. Plot the Phasor diagram of all current and voltage of each elements .

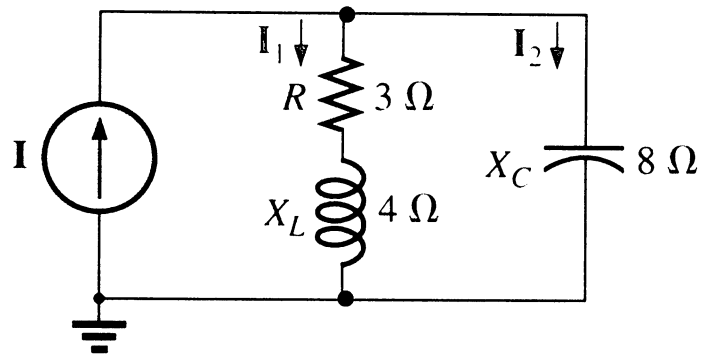


Figure 14

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Instructors: Assoc. Prof. Dr. Titipong Lertwiriaprapa

Dr. Kittisak Phaebua

Answer Sheet

Name: _____ ID: _____

Subject: 341152 Electric Circuits II

Part 1

	True	False
1.		
2.		
3.		
4.		
5.		
6.		
7.		
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9.		
10.		
11.		
12.		
13.		
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16.		
17.		
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19.		
20.		

Part 2

	a	b	c	d
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2.				
3.				
4.				
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